

Math 465: Introduction to Combinatorics

Yelena Mandelshtam

yelenam@umich.edu

These slides will be posted on Canvas.

Course management

I am teaching both of the Math 465 sections offered this Fall:

Section 1: Tuesday and Thursday, 2:30-3:50 PM.

Section 2: Tuesday and Thursday, 4:00-5:20 PM.

The best way to communicate with me is by [email](#).

Most of the course management will occur on [Canvas](#).

My [office hours](#) are:

- Tuesday 1:15–2:15 PM
- Wednesday 1:15–2:15 PM
- Friday 1:45–2:45 PM

These are subject to change!

Grading

The grade will be based on:

- homework (20%),
- quizzes (20%), and
- two exams (30% each).

This course will not be graded on a curve, i.e., there are not a set percentage of each grade to be given out.

Final grade cutoffs:

- total score of 90% guarantees a final grade of A- or higher;
- total score of 80% guarantees a final grade of B- or higher;
- total score of 70% guarantees a final grade of C- or higher;
- total score of 60% guarantees a final grade of D-.

Exams and quizzes

Exams (30% each)

There will be two exams, each covering one half of the course. These exams will **not** be cumulative.

Quizzes (20% total)

There will be **weekly** short (10-15 minute) quizzes, testing you on a problem from the previous homework or basic definitions from lecture. The two lowest quiz scores will be dropped.

Homework

There will be approximately 10 problem sets, posted on Canvas.

Homework #1 is released

It will be due on Tuesday, September 8 at 11:59pm.

Homework **answers will have to be justified.**

No justification—no credit.

Collaboration in small groups (up to four people) is allowed—but each case of collaboration has to be explicitly acknowledged on every homework submission.

The purpose of homework is for you to spend time struggling with the material. Use of AI is not only not permitted, but completely counterproductive to the goals of learning. You will not do well on the exams if you do not put the time into doing the homework yourself.

Homework submission and grading

Homework submission and grading will be done via [Gradescope](#).

Create a Gradescope account if you don't have one already.

On each problem set, only some of the problems will be graded.

Grader: TBA.

Late homework will [not](#) be accepted.

The lowest homework score will be dropped in the final calculation.

Gradescope has a built-in mechanism for regrade requests.

Any questions about the administrative aspects of the course?

Content overview

This course introduces the fundamental notions, techniques, and theorems of [enumerative combinatorics](#) and [graph theory](#).

Level: undergraduate. Next level courses: [Math 565](#), [Math 566](#).

Recommended texts (neither is required):

- M. Bóna, *A walk through combinatorics*, 4th ed., World Scientific, 2016-17.
- S. Shahriari, *Invitation to Combinatorics*, Cambridge University Press, 2022.

We will roughly cover Part II ([Enumerative Combinatorics](#)) and Part III ([Graph Theory](#)) of Bóna's textbook, plus some other topics.

Proofs

Critical skills required: problem solving and proof writing.

This is a **proof-based** course. Students are expected to understand rigorous mathematical proofs, and perhaps even more importantly, supply their own proofs when asked.

Some specific prerequisites:

- proofs by induction,
- proofs by contradiction,
- basic set theory terminology (intersection, union, etc.),
- basic function terminology (one-to-one, onto, composition, etc.).

If you are not familiar with Mathematical Induction or would like a refresher, read either [Bóna, Chapter 2] or [Shahriari, Sections 1.1–1.2].

Problem-solving challenges

In past years, many students ended up dropping out of Math 465.

Why? Much of the content is quite elementary.

However, many problems are “one of a kind,” with no off-the-shelf algorithmic solution.

The quizzes and homework (if you do it without AI) will give you some idea of where you stand.

How do I prepare for quizzes/exams/homework?

practice!

Also, come to office hours.

Questions?

Basic counting principles

First half of the course: [enumerative combinatorics](#).

We will study the basic techniques of [counting](#), or [enumeration](#).

Two versions of counting

- (1) Counting concrete objects;
- (2) Counting choices/possibilities.

Examples

- (1) counting students enrolled in math classes at UM;
- (2) choosing a class schedule for Fall semester.

The golden rule

The golden rule of counting

Count everything **exactly once**.

That is: count everything, and don't count anything more than once.

As we develop more sophisticated methods of counting, it will be important to make mental checks that everything is accounted for, and nothing is overcounted.

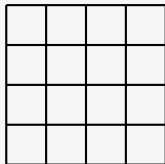
The Addition Principle (=Summation Rule)

Two incarnations of the Addition Principle:

- (1) splitting a set into subsets;
- (2) splitting choices into cases.

Problem

How many squares can be found in this picture?



Solution

$$16 + 9 + 4 + 1 = \mathbf{30}.$$

(We shall later solve this for the $n \times n$ grid.)

Unordered set partitions

Problem

In how many ways can a 4-element set be partitioned into nonempty subsets (=blocks)?

Example: students working in groups on homework.

Solution

There are 4 cases, depending on the number of blocks:

- 1 block: $abcd$
- 2 blocks: $abc|d$ $abd|c$ $acd|b$ $bcd|a$ $ab|cd$ $ac|bd$ $ad|bc$
- 3 blocks: $ab|c|d$ $ac|b|d$ $ad|b|c$ $bc|a|d$ $bd|a|c$ $cd|a|b$
- 4 blocks: $a|b|c|d$

Total: $1+7+6+1=15$.

Exercise: Solve the analogous problem for a 5-element set.

The Multiplication Principle (=Product Rule)

Problem

Falafel can be ordered:

- in a **pita**, a **wrap**, or as a **bowl**;
- with a **regular**, **extra**, or **small** amount of falafel;
- with a **regular**, **extra**, **small**, or **no** amount of hummus;
- with a **regular**, **extra**, **small**, or **no** amount of tahini sauce;
- **with**, **without**, or with **extra** hot sauce;
- **with** or **without** pickles.

How many ways are there to order the *Falafel*?

Solution

$$3 \cdot 3 \cdot 4 \cdot 4 \cdot 3 \cdot 2 = 864.$$

The Multiplication Principle, continued

Problem

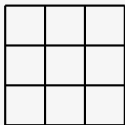
How many divisors does the number 600 have?

Solution

Since $600 = 2^3 \cdot 3 \cdot 5^2$, the answer is $4 \cdot 2 \cdot 3 = 24$.

Problem

How many rectangles are there in this picture?



Solution

$$(3 + 2 + 1) \times (3 + 2 + 1) = \mathbf{36}.$$

Counting permutations

Theorem

An n -element set has $n!$ permutations.

$n = 3$	123	132	213	231	312	321
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Proof #1: selecting the entries, left to right

n choices for the 1st entry, then $n - 1$ choices for the 2nd entry, etc.

Proof #2: inserting the entries, in increasing order

$n = 3$	123	132	213	231	312	321
$n = 4$	1234	1324	2134	2314	3124	3214
	1243	1342	2143	2341	3142	3241
	1423	1432	2413	2431	3412	3421
	4123	4132	4213	4231	4312	4321

Permutations of subsets

Problem

Given 34 people, how many ways are there for at most 4 of them to wait in line?

Solution

Let us divide the set of possibilities into *cases*:

- The line has 0 people. There is 1 way this can happen.
- The line has 1 person. There are 34 ways this can happen.
- The line has 2 people. There are $34 \cdot 33 = 1122$ ways.
- The line has 3 people. There are $34 \cdot 33 \cdot 32 = 35904$ ways.
- The line has 4 people. There are $34 \cdot 33 \cdot 32 \cdot 31 = 1113024$ ways.

Hence the total number of ways is

$$1 + 34 + 1122 + 35904 + 1113024 = \mathbf{1150085}.$$

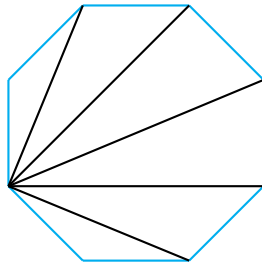
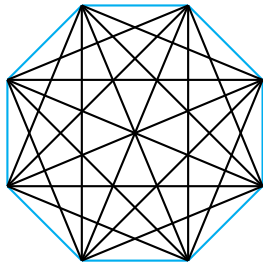
Note that we used both Addition and Multiplication Principles.

The Division Principle

The Division Principle is used in situations where each object of interest is counted several times (a fixed number).

Problem

How many diagonals are there in a convex n -gon?



Solution

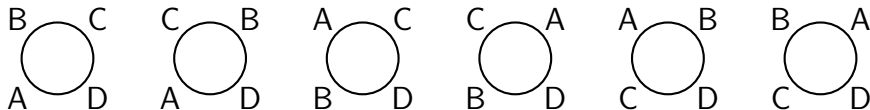
For each of the n vertices, there are $n-3$ diagonals connecting to it. Each diagonal is counted twice. **Answer:** $n(n-3)/2$.

Cyclic permutations

Problem

How many ways are there to arrange n distinct objects in a circle, up to rotation?

Example: seating a party of n people around a round table.



Answer: $(n - 1)!$.

Exercise

Given n beads of different colors, how many circular necklaces are there that use all of these beads?

The Subtraction Principle

The Subtraction Principle is used in situations where it is more practical to count the objects of interest together with some other objects, and then subtract the number of those extra objects.

Problem

How many ways are there to place 5 people named
Amy, Bernie, Cory, Donald, and Elizabeth
on the stage so that Amy and Bernie are not adjacent to each other?

Solution

$$5! - 4! \cdot 2 = 72.$$

Counting words

An **alphabet** is a finite set of symbols (**letters**).

A **word** in an alphabet is a string (i.e., finite sequence) of letters.

Theorem

The number of k -letter words in an n -letter alphabet is n^k .

$k = 2$ $n = 3$	AA	AB	AC	BA	BB	BC	CA	CB	CC
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$k = 3$ $n = 2$	AAA	AAB	ABA	ABB	BAA	BAB	BBA	BBB
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Subsets and binary strings

Theorem

A k -element set has 2^k subsets.

$$\boxed{k = 3} \quad \emptyset \quad \{A\} \quad \{B\} \quad \{C\} \quad \{A,B\} \quad \{A,C\} \quad \{B,C\} \quad \{A,B,C\}$$

Proof

The subsets of a k -element set are encoded by binary strings of length k , i.e., by k -letter words in a 2-letter alphabet:

$$\begin{array}{cccccccc} \emptyset & \{A\} & \{B\} & \{C\} & \{A,B\} & \{A,C\} & \{B,C\} & \{A,B,C\} \\ 000 & 100 & 010 & 001 & 110 & 101 & 011 & 111 \end{array}$$

The number of such words is 2^k .

Boolean functions

Problem

How many Boolean functions in k variables are there?

Solution

Each of the k input bits can take value 0 or 1. There are 2^k possible input binary strings. A Boolean function can send each of them to either 0 or 1. So the total number of possibilities is 2^{2^k} .

Exercise

Count microchips with k binary inputs and ℓ binary outputs. Two microchips are considered identical if they compute the same function $\{\text{binary strings of length } k\} \rightarrow \{\text{binary strings of length } \ell\}$.

For $\ell = 1$, you should recover the previous answer.

Counting words, continued

Theorem

For an n -element alphabet, there are

$$(n)_k \stackrel{\text{def}}{=} n(n-1) \cdots (n-k+1) = \frac{n!}{(n-k)!}$$

words of length k without repetitions.

These numbers are called *falling factorials*.

Proof 1

Use the Multiplication Principle.

Proof 2

Use the Division Principle: take a permutation of the alphabet, then remove the last $n - k$ symbols.

Multinomial coefficients

Theorem

For an alphabet $\{x_1, \dots, x_j\}$ consisting of j symbols, there are

$$\binom{n}{n_1 \dots n_j} \stackrel{\text{def}}{=} \frac{n!}{n_1! \dots n_j!}$$

words of length n consisting of

n_1 copies of letter x_1 ,

.....,

n_j copies of letter x_j .

(Here $n = n_1 + \dots + n_j$.)

Proof

Use the Division Principle: permute n cards carrying those letters.

The numbers $\binom{n}{n_1 \dots n_j}$ are called *multinomial coefficients*.

A multinomial coefficient counts permutations of a *multiset*.

Anagrams

Problem

How many anagrams are there of the word STATISTICS?

Answer

$$\frac{10!}{1! \cdot 1! \cdot 2! \cdot 3! \cdot 3!} = 50400.$$

Problem

How many binary strings consist of three 0's and five 1's?

Answer

$$\frac{8!}{3! \cdot 5!} = 56.$$